Bayesian Classification with Local Probabilistic Model Assumption in Aiding Medical Diagnosis

Bin Hu*, Chengsheng Mao, Xiaowei Zhang, Yongqiang Dai
School of Information Science and Engineering, Lanzhou University, Lanzhou, China
*Email: bh@lzu.edu.cn

***Abstract*—In computer-aided diagnosis, a Bayesian classifier that can give the class membership probabilities should be more favorable than classifiers that only give a class assertion. In Bayesian classification, an important and critical step is the probability distribution estimation for each class. Existing methods usually estimate the probability distribution in the whole sample space where the original distribution may be too complex to model. In this paper, we propose a probability distribution estimation method based on local probabilistic model assumption. In our method, the estimation of global probability for a certain point is transformed to the computation of local distribution in a small region, where the local distribution is supposed to be simpler and can be assumed as a simpler probabilistic model. By this method, we implement the Bayesian classifiers based on several local probabilistic model assumptions, and experiments with these classifier have been conducted on several real-word biological and medical datasets; the experimental results demonstrate the efficacy of the proposed method for probabilistic classification in medical diagnosis.**

***Keywords*—Bayesian classification; probabilistic model; computer-aided diagnosis; local learning.**

I. Introduction

In medical diagnosis, a classifier is usually modelled to assist doctors in getting an accurate diagnosis. For example, a forward neural network was applied as a classifier to identify abnormal brain from normal controls in [1], and encouraging results are achieved. Also, Zhang et.al [2] proposed using kernel support vector machine decision tree as a classifier to identify subjects with Alzheimer's and mild cognitive impairment from normal elder controls. Although a common classifier can give a classification result, it may not give the confidence level of the identification, which can be referred to by the doctors to give an accurate diagnosis.

A Bayesian classifier can predict the class membership probabilities based on Bayes' Theory. A membership probability can be viewed as a degree of confidence for the diagnosis; for example, the probability of a patient having cancer should be more meaningful than a simple assertion. Due to the output of class membership probabilities, Bayesian classifiers can be a good choice for Computer-Aided Diagnosis (CAD).

Researchers have developed several methods in estimating the probability distribution in Bayesian classification. The Naive Bayesian classifier (NBC) takes the class conditional independence assumption to simplify the computation of probability distribution, and can even exhibit high accuracy under certain conditions [3]. However, the class conditional independence assumption is rarely true in real-world applications. The Bayesian belief networks (BBN) [4] were proposed to allow the representation of dependencies among the features, but it is shown that the general problem of inference in unconstrained belief networks is NP-hard [5], [6]. The common BBN (e.g. TAN [4] and AODE [7]) usually relax the conditional independence to make a compromise between the classification effectiveness and efficiency. The non-naive Bayesian classifier (NNBC) [8] estimates the joint probability density function (PDF) using multivariate kernel density estimation (KDE) with a Parzen window method; however, due to the complex Gaussian mixture distribution assumption in NNBC, the NNBC is easy to cause the overfitting issue.

In this paper, we research the Bayesian classification method from the perspective of local probabilistic model and propose a novel Bayesian Classification method with Local Model Assumption (LMA-BC). In the proposed method, the estimation of global probability distribution is transformed to estimating the local probability distribution in a small local region where the probability distribution is supposed to be simple and can be estimated through assuming a simple local probabilistic model. By using a local model assumption for a certain sample, accurate probabilities can be estimated effectively and efficiently while relaxing the class conditional independent assumption. As LMA-BC can output the posterior probability effectively and efficiently, it would be a favorable classifier for CAD.

II. Classification with LMA

A. Local Distribution

We use local distribution to describe the distribution of a variable[1] in a small local region, versus the global distribution widely used in previous research.

Similar to PDF for the whole sample space, we use local probability density (LPD) to describe the local distribution in a certain region. In the sample space of a variable $\mathbf{x}$, given a continuous closed region R, for an arbitrary point $\mathbf{x}$, the LPD in R, denoted as $f_R(\mathbf{x})$, is defined as

$$f_R(\mathbf{x}) = \lim_{\delta(\mathbf{x}) \to \mathbf{x}} \frac{P(\delta(\mathbf{x})|R)}{V(\delta(\mathbf{x}))} \tag{1}$$

where $\delta(\mathbf{x})$ and $V(\delta(\mathbf{x}))$ respectively denote the neighborhood of $\mathbf{x}$ and the its volume, and $P(\delta(\mathbf{x})|R)$ is the conditional probability that a point is in $\delta(\mathbf{x})$ given that it is in R.

[1]It can be a single variable for univariate distribution or a multi-variable vector for multivariate joint distribution, same as below.

LPD has some similar probabilities with PDF. It also has the nonnegativity and unitarity in the corresponding region. With LPD, the global probability density of point in local region R can be computed out as

$$f(\mathbf{x}) = f_R(\mathbf{x})P(R) \quad for \quad \mathbf{x} \in R. \tag{2}$$

It provides a method to estimate PDF from LPD, which is the basic idea of our method. Given an arbitrary point $\mathbf{x}$ in the sample space, we consider a local region R containing $\mathbf{x}$; if the prior probability $P(R)$ is known or can be estimated, the estimation of PDF is transformed to the estimation of LPD.

B. LPD Estimation

If there are k samples $\mathbf{x}_1, \cdots, \mathbf{x}_k$ in n observations falling into the local region R, the prior probability $P(R)$ is usually estimated as $\hat{P}(R) = k/n$. As for the estimation of LPD, we take a local model assumption (LMA) that a probabilistic model is assumed in the local region. It is similar to the estimation of PDF in the whole sample space, except that the local model is estimated based on the samples falling in the corresponding local region.

1) Locally Uniform Assumption: If a local region R is sufficiently small, the region can simply be assumed uniform. This is the Locally Uniform Assumption (LUA). According to the unitarity of LPD in region R, the LPD with LUA is estimated as

$$\hat{f}_R(\mathbf{x}) = \frac{1}{V(R)}, \quad for \quad \mathbf{x} \in R. \tag{3}$$

With the prior probability of R usually estimated by $\hat{P}(R) = k/n$, the PDF can be estimated as $\hat{f}(\mathbf{x}) = \frac{k}{nV(R)}$, in agreement with the histogram estimator.

However, the local region can not always be selected sufficiently small. If the sample size n are not large enough, to achieve a more accurate estimation for $P(R)$, the local region R may be somewhat large and can not be always regarded uniform.

2) Locally Parametric Assumption: If the local region can not be small enough, we may assume a parametric model to model the local distribution. This is the Locally Parametric Assumption (LPA) where we assumes samples in the local region R follow a known distribution $f(\mathbf{x}; \theta)$ with an unknown parameter vector θ. With the samples in R, an estimate of θ can be obtained. Then $f(\mathbf{x}; \hat{\theta})$ can be used to describe the local distribution in R. Due to the unitarity, the LPD in R should be normalized. Specially, a Gaussian model with unknown mean and covariance matrix can also be assumed in a local region; we call this locally Gaussian assumption (LGA).

3) Locally Complex Assumption: If the local region is too large, simple parametric model can not model the local distribution, we may take the locally complex assumption (LCA) and employ a non-parametric method to make an accurate estimation. Similarly, the KDE can also be employed for the LPD estimation with an assigned bandwidth vector $\mathbf{h}$. LCA estimates the local distribution from the samples in local region R as

$$\hat{f}(\mathbf{x}) = \frac{1}{k \cdot prod(\mathbf{h})} \sum_{i=1}^{k} K((\mathbf{x} - \mathbf{x}_i)./\mathbf{h}) \tag{4}$$

where the operator ./ denotes the right division between the corresponding elements in two equal-sized matrices or vectors; $prod(\cdot)$ returns the product of the all elements in a vector; and $K(\cdot)$ is the kernel function. The estimation of LPD also need to be normalized.

In terms of the complexities of the local models, LUA is the simplest, LCA is the most complex and LPA contains a series of parametric model assumptions whose complexity is between LUA and LCA. Generally, the selection of LMA depends on the corresponding local region; if the local region is small and has only a few samples in it, a simpler LMA is favored; if a complex model is assumed in a small local region, it will cause the overfitting issue.

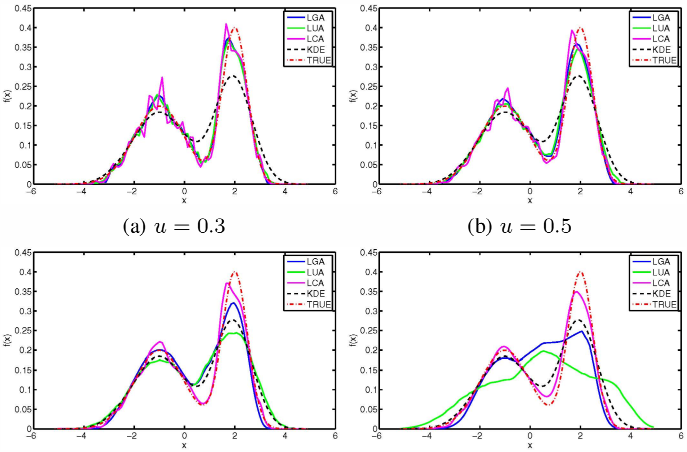
Usually, for a certain point $\mathbf{x}$, its neighborhood $\delta_u = \{\mathbf{y} : d(\mathbf{x}, \mathbf{y}) \leq u\}$ is selected as the local region to estimate the global probability density; u denotes the neighborhood size. We have tested the performance of LPD estimation in one-dimensional cases where the neighborhood of x is an interval $[x-u, x+u]$. 1000 samples have been selected from a Gaussian mixture distribution, and then we estimate the probability density from these samples by LMA and KDE. Since the local regions for different points may not be disjunctive, the PDF computed by LPD will not be guaranteed to integrated to unity, we make a normalization for the probability densities. The estimated PDF curves for different neighborhood sizes and LMAs are shown in Figure 1. Figure 1a shows that taking a LCA in a small neighborhood size ($u = 0.3$) usually cause the overfitting issue and the corresponding estimated curve is quite unsmooth; a simpler LMA like LUA and LGA is more appropriate. Figure 1d shows that in a large enough neighborhood ($u = 2$) LCA can be more effective than LUA and LGA. Figure 1b and 1c show that in a Medium-sized neighborhood ($u = 0.5$ and $u = 1$), LGA can be more effective. Generally, with the neighborhood expanding, a more complex LMA should be taken.

C. Classification Rules

For a certain $\mathbf{x}$, LMA-BC takes its neighborhood $\delta_i(\mathbf{x})$ as the local region to estimate the LPD and then PDF for the ith class y_i of which the sample set is denoted by T_i. We use $T_{\delta_i(\mathbf{x})}$ to denote the set of samples from T_i falling in $\delta_i(\mathbf{x})$, i.e. $T_{\delta_i(\mathbf{x})} = T_i \bigcap \delta_i(\mathbf{x})$. The prior probability of $\delta_i(\mathbf{x})$ for class y_i can be estimated as $\hat{P}(\delta_i(\mathbf{x})) = |T_{\delta_i(\mathbf{x})}|/|T_i|$. Then according to the Bayesian rule, the optimal class by LMA-BC is $y_{\hat{w}}$, with

$$\begin{aligned} \hat{w} &= \max_{i=1,\cdots,c} P(y_i|\mathbf{x}) \\ &= \max_{i=1,\cdots,c} P(\mathbf{x}|y_i)P(y_i)/P(\mathbf{x}) \\ &= \max_{i=1,\cdots,c} \frac{|T_i|}{n} \cdot \frac{|T_{\delta_i(\mathbf{x})}|}{|T_i|} \cdot f_{\delta_i(\mathbf{x})}(\mathbf{x}) \\ &= \max_{i=1,\cdots,c} |T_{\delta_i(\mathbf{x})}| \cdot f_{\delta_i(\mathbf{x})}(\mathbf{x}) \end{aligned} \tag{5}$$

where $f_{\delta_i(\mathbf{x})}(\mathbf{x})$ is the LPD for point $\mathbf{x}$ in $\delta_i(\mathbf{x})$.



(a) $u = 0.3$ (b) $u = 0.5$ (c) $u = 1$ (d) $u = 2$

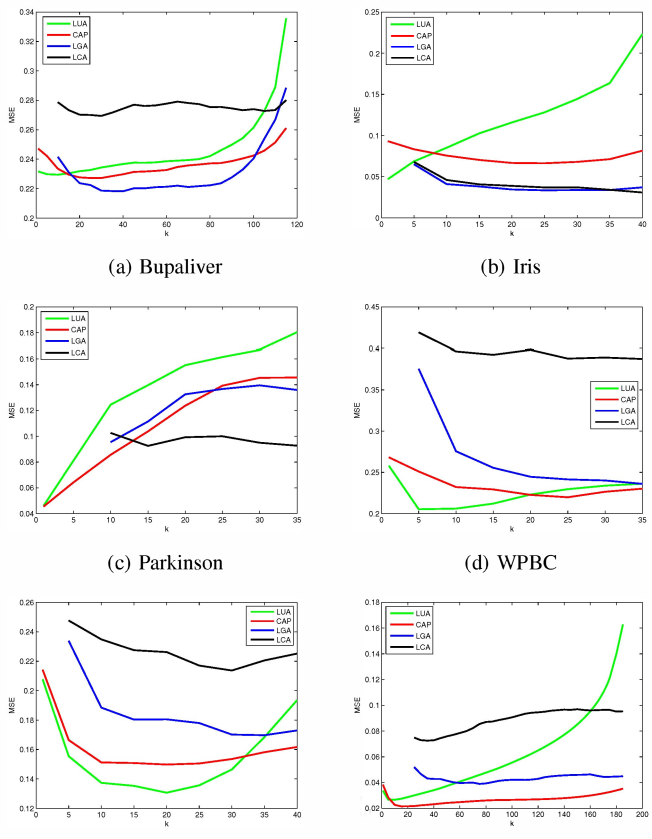
Fig. 1: An example of estimating the probability density in one-dimensional case by local distribution with various neighborhood size. The true distribution is $f(x) = 0.5g(x+1) + 0.5g(\frac{x-2}{0.5})$, where $g(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$ is the standard Gaussian function.

As a probabilistic classifier, LMA-BC outputs the posterior probability as

$$\hat{P}(y_i|\mathbf{x}) = \frac{|T_{\delta_i(\mathbf{x})}| \cdot f_{\delta_i(\mathbf{x})}(\mathbf{x})}{\sum_{j=1}^{c} |T_{\delta_j(\mathbf{x})}| \cdot f_{\delta_j(\mathbf{x})}(\mathbf{x})} \tag{6}$$

From Formula 5, LMA-BC can be considered as a generalization of some existing classification methods through varying the local model assumption and the corresponding local region. Theoretically, an arbitrary neighborhood can be selected for LMA-BC. Nonetheless, some rules of neighborhood selection may be taken to raise effectiveness and efficiency. A regular neighborhood should be selected to make it more efficient to judge whether a sample is in the neighborhood and to compute the volume or integration in the neighborhood. Usually, a neighborhood centered at $\mathbf{x}$ is favored.

In addition, some rules can be taken so that the neighborhood between classes have certain common characteristics. Usually, two strategies can be used to select neighborhoods between classes: 1) Select the same neighborhood for all the classes; with the same neighborhood $\delta(\mathbf{x}; r)$, the LPD for each class is estimated in the same local region. If taking the LUA, the estimates of LPD is equal among classes, and LMA-BC outputs the class with maximum $|T_i \bigcap \delta(\mathbf{x}; r)|$, i.e. the most common class in the neighborhood, in agreement with the well-known K-Nearest-Neighbor (KNN) rules [9]. 2) Select a regular neighborhood for each class that each neighborhood has an equal number of neighbors in the corresponding class, i.e. $|T_{\delta_i(\mathbf{x})}|$ is equal for all classes. With this strategy, LMA-BC assigns the class with the greatest LPD to $\mathbf{x}$; for LUA the class with the nearest kth sample is assigned, it is actually a kNN rule with a certain neighborhood size. Some similar methods can also be considered a special case of LMA-BC in this case. For example, the categorical-average-pattern (CAP) method [10] and the local mean-based classifier [11] minimize the distance between the query sample and the local centers, which are special cases of LMA-BC taking the LGA with an identity covariance matrix.



(a) Bupaliver (b) Iris (c) Parkinson (d) WPBC (e) Spectf (f) WBC

Fig. 2: Variation of MSE for different LMA with the neighborhood size.

III. Experimental Evaluation

To evaluate the performance of LMA-BC, experiments are performed on six benchmark datasets about medical diagnosis from the UCI machine learning repository [12]. Detailed information is summarized on the left side in Table I.

In our experiments, each dataset is randomly stratified into 5 folds; 4 folds constitute the training set, the remaining fold is the test set. The mean square error (MSE) and the classification accuracy (ACC) on the test set are used to evaluate the performance of a classifier. To avoid bias, the 5-fold cross validation is applied to each dataset 10 times and the performance is averaged. In addition, before classification the features are normalized over the training data to have zero center and unit variance, and the test data features are normalized using the corresponding training center and variance.

A. Variation with Neighborhood Size and LMA

The neighborhood size and LMA can affect the estimation of probability distribution and then affect classification performance. In our experiments, for a test data $\mathbf{x}$, the neighborhood for the ith class T_i is selected as $\delta_i(\mathbf{x}, k) = \{\mathbf{z} : d(\mathbf{x}, \mathbf{z}) \leq d_k^{(i)}\}$ where $d_k^{(i)}$ is the distance between the kth nearest neighbor in T_i and $\mathbf{x}$, and the distance metric $d(\cdot)$ is Euclidean distance. We vary k which can indicate the neighborhood size, and take different LMA to test the performances of LMA-BC; the variations of the MSE on these datasets are shown

TABLE I: The dataset information and the performance of different probabilistic classifiers in terms of MSE and ACC. The best and comparable results (according to t-test with significance at 0.05) are highlighted.

Datasets	#Samples	#Features	#Classes		LUA-BC	CAP-BC	LGA-BC	LCA-BC	LR	GMA-NBC	KDE-NBC	NNBC	TAN
Bupaliver	345	6	2	MSE	0.2298	0.2335	**0.2168**	0.2743	**0.2139**	0.2614	0.2237	0.2505	0.2443
				ACC	0.6284	0.6513	**0.6794**	0.6490	0.6733	0.5580	0.6461	0.6128	0.5768
Iris	150	4	3	MSE	0.0686	0.0756	0.0337	0.0308	**0.0221**	0.0370	0.0349	0.0454	0.0345
				ACC	0.9533	0.9560	0.9600	**0.9620**	0.9533	0.9520	0.9587	0.9447	0.9267
Spectf	267	44	2	MSE	**0.1554**	**0.1512**	0.1702	0.2252	0.1732	0.3006	0.2440	0.2793	**0.1586**
				ACC	0.7640	0.7764	**0.8048**	0.7652	0.7767	0.6832	0.7383	0.6981	**0.7903**
WPBC	198	32	2	MSE	**0.2052**	0.2321	0.2401	0.3875	0.2248	0.3016	0.3192	0.2953	**0.1865**
				ACC	**0.7583**	0.7171	**0.7531**	0.5872	0.7006	0.6356	0.6178	0.6268	**0.7629**
Parkinsons	195	22	2	MSE	**0.0789**	**0.0842**	0.1398	0.0864	0.1225	0.2873	0.2402	0.1718	0.1251
				ACC	0.8865	**0.8958**	0.8537	**0.9031**	0.8459	0.7025	0.7487	0.8082	0.8359
WBC	683	9	2	MSE	0.0267	**0.0229**	0.0471	0.0720	0.0259	0.0366	0.0282	0.0362	0.0241
				ACC	0.9695	**0.9748**	0.9485	0.9237	0.9663	0.9618	0.9672	0.9617	**0.9707**

in Figure 2.[2] It can be seen from Figure 2 that a smaller neighborhood favors a simpler LMA, as expected; while in a larger neighborhood, a more complex LMA usually performs better. However, a complex LMA is not always perform better than a simple one even in large neighborhood; it depends on the natural complexity of the true classification problem. In the practical application, we can choose an appropriate LMA and neighborhood to achieve an accurate diagnosis.

B. Comparison with Other Classifiers

Based on the variation of neighborhood size with LMA, an optimal neighborhood can be selected to adapt to an LMA. For each dataset, we implement four types of LMA-BC including LUA, CAP, LGA and LCA, and the corresponding neighborhood size is selected as $k = 5$, $k = 10$, $k = 30$, and $k = 40$ respectively. In addition, five other naturally probabilistic classifiers are implemented to validate the performance of LMA-BC, including the basic NBC with GMA, the flexible NBC with KDE [13], a sophisticated probabilistic classifier logistic regression (LR) [14], a recent non-naive classifier NNBC [8] and a Bayesian network TAN. The results of different classifiers on various datasets are reported in Table I. As can be seen, Bayesian classifiers with a certain LMA and the corresponding neighborhood size can be comparable to or outperform other competitors in most cases in terms of MSE and ACC.

IV. Discussion and Conclusion

In this paper, we proposed a probability distribution estimation method with local model assumption for Bayesian classification in medical diagnosis. In our method, the estimation of global distribution is transformed to the estimation of distribution in a local region. In addition, we implemented a Bayesian classifier based on the LMA. As a probabilistic classifier, LMA-BC can output the membership probability for a further diagnosis. Different classifiers may generate the same classification rule but output different membership probabilities. Experiments on several real-world datasets validate the effectiveness of the proposed Bayesian classifier for medical diagnosis.

[2]The covariance matrix of LGA is assumed diagonal. CAP is a LGA with identical covariance matrix, the complexity relationship is: LUA $<$CAP $<$LGA $<$LCA. LGA and LCA are invalid in a very small neighborhood, so the corresponding MSE is null. The kernel in LCA is Gaussian.

Acknowledgment

This work was supported by the National Basic Research Program of China (2014CB744600), the National Natural Science Foundation of China (61402211, 61063028 and 61210010) and the International Cooperation Project of Ministry of Science and Technology (2013DFA11140).

References

[1] Y. Zhang, S. Wang, G. Ji, and Z. Dong, "Genetic pattern search and its application to brain image classification," *Mathematical Problems in Engineering*, vol. 2013, 2013.

[2] Y. Zhang, S. Wang, and Z. Dong, "Classification of alzheimer disease based on structural magnetic resonance imaging by kernel support vector machine decision tree," *Progress In Electromagnetics Research*, vol. 144, pp. 171–184, 2014.

[3] P. Domingos and M. Pazzani, "Beyond independence: Conditions for the optimality of the simple bayesian classi er," in *Proc. 13th Intl. Conf. Machine Learning*, 1996, pp. 105–112.

[4] N. Friedman, D. Geiger, and M. Goldszmidt, "Bayesian network classifiers," *Machine learning*, vol. 29, no. 2-3, pp. 131–163, 1997.

[5] G. F. Cooper, "The computational complexity of probabilistic inference using bayesian belief networks," *Artificial intelligence*, vol. 42, no. 2, pp. 393–405, 1990.

[6] K. B. Laskey and S. M. Mahoney, "Network fragments: Representing knowledge for constructing probabilistic models," in *Proceedings of the Thirteenth conference on Uncertainty in artificial intelligence*. Morgan Kaufmann Publishers Inc., 1997, pp. 334–341.

[7] G. I. Webb, J. R. Boughton, and Z. Wang, "Not so naive bayes: aggregating one-dependence estimators," *Machine learning*, vol. 58, no. 1, pp. 5–24, 2005.

[8] X.-Z. Wang, Y.-L. He, and D. D. Wang, "Non-naive bayesian classifiers for classification problems with continuous attributes," *Cybernetics, IEEE Transactions on*, vol. 44, no. 1, pp. 21–39, 2014.

[9] D. T. Larose and C. D. Larose, "k-nearest neighbor algorithm," *Discovering Knowledge in Data: An Introduction to Data Mining, Second Edition*, pp. 149–164, 2006.

[10] S. Hotta, S. Kiyasu, and S. Miyahara, "Pattern recognition using average patterns of categorical k-nearest neighbors," in *Pattern Recognition, 2004. ICPR 2004. Proceedings of the 17th International Conference on*, vol. 4. IEEE, 2004, pp. 412–415.

[11] Y. Mitani and Y. Hamamoto, "A local mean-based nonparametric classifier," *Pattern Recognition Letters*, vol. 27, no. 10, pp. 1151–1159, 2006.

[12] K. Bache and M. Lichman, "UCI machine learning repository," 2013. [Online]. Available: http://archive.ics.uci.edu/ml

[13] G. H. John and P. Langley, "Estimating continuous distributions in bayesian classifiers," in *Proceedings of the Eleventh conference on Uncertainty in artificial intelligence*. Morgan Kaufmann Publishers Inc., 1995, pp. 338–345.

[14] D. A. Freedman, *Statistical models: theory and practice*. cambridge university press, 2009.